A FINITELY PRESENTED MAP OF AFFINE SCHEMES DESCENDS TO A FINITE TYPE MAP OVER A Z-SCHEME OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9738-9754 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9756-9764, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. This is the parent approximation lemma of the chapter's Noetherian-approximation family; module (L16009), flat-module (L16003) and smooth (L16045) upgrades are already registered separately with distinct objects of approximation.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The characterization of finite presentation of morphisms of affines by finite presentation of the corresponding algebra map (Morphisms, lemma-locally-finite-presentation-characterize) and the algebra lemma presenting a finite presentation algebra over a filtered colimit at a finite stage (Algebra, lemma-limit-module-finite-presentation); cited inputs with the translation written in the retained span.

Source reference. Original lemma 00R1: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9764–9764. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01TQ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9764–9764. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of affine schemes, which is of finite presentation. Then there exists a cartesian diagram*

$$\begin{array}{ccc} X_0 & \xleftarrow{g} & X \\ {\scriptstyle f_0}\downarrow & & \downarrow{\scriptstyle f} \\ S_0 & \longleftarrow & S \end{array}$$

such that

(1) *X_0, S_0 are affine schemes,*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(2) S_0 *is of finite type over* $\mathbf{Z}$,
(3) f_0 *is of finite type.*

Proof. Write $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$. As f is of finite presentation we see that B is of finite presentation as an A-algebra, see Morphisms, Lemma 01TQ. Thus the lemma follows from Algebra, Lemma 00R1. □